

# Math 3A03 Handout

## *The (Volterra) Integral Operator $T$ on $C[a, b]$ , Continuity of $T$ wrt Standard $p$ -Norms, and Failure for More Exotic Structures*

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This handout gives details about a few issues discussed in class, and digs a bit deeper. I prepared the handout by prompting ChatGPT to write a latex document, iterating until I had most of the content of the document I wanted, and then editing for a couple of hours on my own.

–David Earn

## Introduction

Let

$$C[a, b] := \{f : [a, b] \rightarrow \mathbb{R} : f \text{ is continuous}\}$$

be the vector space of all real-valued continuous functions on the closed interval  $[a, b]$ .

In this handout we study the *Volterra integral operator*

$$(Tf)(x) := \int_a^x f(t) dt.$$

This is one of the simplest and most important examples of an integral operator: it assigns to each function  $f$  the accumulated area under the graph of  $f$  from  $a$  to  $x$ .

Our goals are:

1. to prove that  $T$  is continuous with respect to every usual  $p$ -norm on  $C[a, b]$ ;
2. to explain that if one allows arbitrary metrics, then producing a metric for which  $T$  is not continuous is very easy;

3. to show that even if one insists on a *norm*, there still exists a norm on  $C[a, b]$  for which  $T$  is not continuous;
4. to explain the algebraic idea behind that norm, namely a *Hamel basis*.

Result 1 is standard analysis. Since  $\int(\alpha f + \beta g) = \alpha \int f + \beta \int g$ , points 2 and 3 imply that a linear operator on  $C[a, b]$  need not be continuous (which what you might guess for something that is “linear”). Point 4 is a good illustration of how infinite-dimensional vector spaces can behave in ways that are very different from finite-dimensional spaces.

## 1 The usual $p$ -norms on $C[a, b]$

For  $1 \leq p < \infty$ , define

$$\|f\|_p := \left( \int_a^b |f(x)|^p dx \right)^{1/p},$$

and define the sup norm by

$$\|f\|_\infty := \max_{x \in [a, b]} |f(x)|.$$

Because  $f$  is continuous on a closed interval, all of these quantities are finite.

The norm  $\|\cdot\|_\infty$  measures the largest value of  $|f|$ , while the norms  $\|\cdot\|_p$  for  $1 \leq p < \infty$  measure the average size of  $|f|^p$ .

## 2 The Volterra operator

**Definition 1.** The Volterra integral operator is the map  $T : C[a, b] \rightarrow C[a, b]$  defined by

$$(Tf)(x) := \int_a^x f(t) dt.$$

If  $f \in C[a, b]$ , then  $Tf$  is again continuous (by the “integrals are continuous” theorem), and in fact the Fundamental Theorem of Calculus tells us that  $Tf$  is differentiable and

$$(Tf)'(x) = f(x).$$

## 3 Continuity of $T$ with respect to every $p$ -norm

We first record a standard fact about linear operators between normed spaces.

**Lemma 1.** Let  $X$  and  $Y$  be normed vector spaces, and let  $S : X \rightarrow Y$  be linear. If there exists a constant  $K \geq 0$  such that

$$\|Sf\|_Y \leq K \|f\|_X \quad \text{for all } f \in X,$$

then  $S$  is continuous.

*Proof.* To prove continuity at an arbitrary point  $f_0 \in X$ , let  $\varepsilon > 0$  be given and choose

$$\delta := \frac{\varepsilon}{K+1}.$$

If  $\|f - f_0\|_X < \delta$ , then by linearity,

$$\|Sf - Sf_0\|_Y = \|S(f - f_0)\|_Y \leq K \|f - f_0\|_X < K\delta < \varepsilon.$$

Thus  $S$  is continuous at  $f_0$ . Since  $f_0$  was arbitrary,  $S$  is continuous on  $X$ . □

**Remark 1.** *The inequality*

$$\|Sf\|_Y \leq K \|f\|_X \quad \text{for all } f \in X$$

*says that the operator  $S$  can enlarge vectors by at most the fixed factor  $K$ . For this reason, one says that  $S$  is a bounded operator.*

*In more advanced courses in analysis, one defines a norm on the space of linear operators from  $X$  to  $Y$ , namely*

$$\|S\|_{\text{op}} := \sup_{\|f\|_X \leq 1} \|Sf\|_Y,$$

*and this terminology becomes even more natural: the condition above is exactly the statement that this supremum is finite. In other words, a linear operator is called bounded precisely when it has finite operator norm.*

*For linear maps between normed spaces, the important fact is that*

$$\text{boundedness} \iff \text{continuity}.$$

*So in practice, proving an estimate of the form*

$$\|Sf\|_Y \leq K \|f\|_X$$

*is the standard way to prove that a linear operator is continuous.*

We now prove the main analytic fact, exploiting the Lemma above and the fact that the integral is a linear operator.

**Theorem 1.** *For every  $1 \leq p \leq \infty$ , the Volterra operator*

$$T : (C[a, b], \|\cdot\|_p) \rightarrow (C[a, b], \|\cdot\|_p)$$

*is continuous. More precisely,*

$$\|Tf\|_p \leq (b-a) \|f\|_p \quad \text{for all } f \in C[a, b].$$

*Proof.* We treat the cases  $p = \infty$  and  $1 \leq p < \infty$  separately.

First suppose  $p = \infty$ . For each  $x \in [a, b]$ ,

$$|(Tf)(x)| = \left| \int_a^x f(t) dt \right| \leq \int_a^x |f(t)| dt \leq \int_a^x \|f\|_\infty dt \leq (b-a) \|f\|_\infty.$$

Taking the maximum over  $x$  gives

$$\|Tf\|_{\infty} \leq (b-a) \|f\|_{\infty}.$$

Now suppose  $1 \leq p < \infty$ . For each  $x \in [a, b]$ ,

$$|(Tf)(x)| \leq \int_a^x |f(t)| dt \leq \int_a^b |f(t)| dt = \|f\|_1.$$

Hence

$$\|Tf\|_{\infty} \leq \|f\|_1.$$

On the finite interval  $[a, b]$ , Hölder's inequality (see the handout on  $p$ -norms) implies

$$\|f\|_1 \leq (b-a)^{1-1/p} \|f\|_p.$$

Therefore

$$\|Tf\|_{\infty} \leq (b-a)^{1-1/p} \|f\|_p.$$

Using the elementary inequality

$$\|g\|_p \leq (b-a)^{1/p} \|g\|_{\infty} \quad \text{for all } g \in C[a, b],$$

we get

$$\|Tf\|_p \leq (b-a)^{1/p} \|Tf\|_{\infty} \leq (b-a)^{1/p} (b-a)^{1-1/p} \|f\|_p = (b-a) \|f\|_p.$$

This proves the claim for all  $1 \leq p \leq \infty$ . □

**Remark 2.** *The theorem shows that the Volterra operator is continuous for every familiar  $L^p$ -type norm on  $C[a, b]$ . This is why, in ordinary analysis, one thinks of  $T$  as a very well-behaved operator.*

## 4 A simpler question: what if we only ask for a metric?

Before looking for a norm with respect to which  $T$  is not continuous, it is worth observing that if one allows *arbitrary metrics*, the question becomes much easier.

**Definition 2.** *The discrete metric on a set  $X$  is defined by*

$$d(x, y) = \begin{cases} 0, & x = y, \\ 1, & x \neq y. \end{cases}$$

**Proposition 1.** *If the domain  $C[a, b]$  is given its usual sup-norm metric and the codomain  $C[a, b]$  is given the discrete metric, then the Volterra operator  $T$  is not continuous.*

*Proof.* Consider the sequence of constant functions

$$f_n(x) = \frac{1}{n}.$$

Then  $f_n \rightarrow Z$  in the sup norm (where  $Z$  is the zero function,  $Z(x) = 0 \forall x$ ), because

$$\|f_n - Z\|_\infty = \frac{1}{n} \rightarrow 0.$$

However,

$$(Tf_n)(x) = \int_a^x \frac{1}{n} dt = \frac{x-a}{n},$$

which is not the zero function for any  $n$ . Thus, in the discrete metric on the codomain,

$$d(Tf_n, TZ) = 1 \quad \text{for all } n.$$

So  $Tf_n \not\rightarrow TZ$ . Therefore  $T$  is not continuous. □

*Alternative proof using inverse images of open sets.* Let

$$U := \{Z\} \subseteq C[a, b],$$

where the codomain is equipped with the discrete metric. Since every subset of a discrete metric space is open,  $U$  is open.

Now

$$T^{-1}(U) = T^{-1}(\{Z\}) = \{f \in C[a, b] : Tf = Z\}.$$

By the Fundamental Theorem of Calculus, if  $Tf = Z$ , then

$$Z = (Tf)' = f,$$

so  $f = Z$ . Thus

$$T^{-1}(U) = \{Z\}.$$

But  $\{Z\}$  is not open in  $C[a, b]$  with the sup-norm metric: every sup-norm ball centered at  $Z$  contains nonzero constant functions. Therefore the inverse image of the open set  $U$  is not open, so  $T$  is not continuous. □

**Remark 3.** *This example is very simple, but it is not very interesting analytically, because the discrete metric has nothing to do with the vector-space structure of  $C[a, b]$  or the usual analytic structures on it (e.g., the integral operator). The more interesting question is whether there exists a norm on  $C[a, b]$  for which  $T$  is not continuous.*

## 5 Why the norm question is subtler

In finite-dimensional vector spaces, all norms induce the same notion of continuity. Thus, in finite dimensions, if a linear operator is continuous for one norm, then it is continuous for every norm.

In infinite-dimensional spaces this is false. Different norms can give genuinely different topologies, so continuity can depend strongly on the chosen norm.

For the usual norms  $\|\cdot\|_p$ , the Volterra operator is continuous. But there are much more exotic norms on  $C[a, b]$ , and with respect to one of those norms the operator  $T$  can fail to be continuous.

To build such a norm, we need one algebraic idea.

## 6 Hamel bases

**Definition 3.** Let  $V$  be a vector space over  $\mathbb{R}$ . A Hamel basis for  $V$  is a subset  $\mathcal{B} \subseteq V$  such that every vector  $v \in V$  can be written in one and only one way as a finite linear combination of elements of  $\mathcal{B}$ .

This is the same notion of basis that you will have seen in finite-dimensional linear algebra; the only difference is that in an infinite-dimensional vector space the basis itself is infinite.

**Example 1.** In the vector space of polynomials, the set

$$\{1, x, x^2, x^3, \dots\}$$

is a Hamel basis, because every polynomial is a finite linear combination of powers of  $x$ .

For the space  $C[a, b]$ , the existence of a Hamel basis is a standard theorem from abstract linear algebra (usually proved using the axiom of choice). In practice one cannot write such a basis down explicitly, but one can still use it to build examples.

The only fact we need is the following.

**Proposition 2.** Every linearly independent subset of  $C[a, b]$  can be extended to a Hamel basis of  $C[a, b]$ .

We will not prove this here; it is a standard basis extension theorem.

## 7 A linearly independent family adapted to the Volterra operator

Define, for  $n = 0, 1, 2, \dots$ ,

$$e_n(x) := \left( \frac{x-a}{b-a} \right)^{2n}.$$

Then each  $e_n$  belongs to  $C[a, b]$ , and

$$\|e_n\|_\infty = 1.$$

Applying the Volterra operator gives

$$(Te_n)(x) = \int_a^x \left( \frac{t-a}{b-a} \right)^{2n} dt = \frac{b-a}{2n+1} \left( \frac{x-a}{b-a} \right)^{2n+1}.$$

Thus  $e_n$  is an even power of  $\frac{x-a}{b-a}$ , while  $Te_n$  is a constant multiple of an odd power.

**Lemma 2.** *The family*

$$\{e_0, e_1, e_2, \dots\} \cup \{Te_0, Te_1, Te_2, \dots\}$$

*is linearly independent.*

*Proof.* Suppose a finite linear combination is zero:

$$\sum_{k=0}^N \alpha_k e_k + \sum_{k=0}^N \beta_k Te_k = 0.$$

Substituting the formulas for  $e_k$  and  $Te_k$  gives a polynomial identity in the variable

$$y := \frac{x-a}{b-a}.$$

The first sum involves only even powers  $y^{2k}$ , while the second sum involves only odd powers  $y^{2k+1}$ . A polynomial is identically zero only if all its coefficients are zero, so all the  $\alpha_k$  and  $\beta_k$  must vanish. Hence the family is linearly independent.  $\square$

## 8 The general Hamel-basis strategy

The Hamel-basis construction is not tied specifically to the sup norm. In fact, once one fixes any norm  $N$  on  $C[a, b]$ , one can often build a new norm of the form

$$\|f\|_{N,*} := N(f) + |L(f)|$$

for which a given linear operator fails to be continuous.

For the Volterra operator, this works starting from each of the usual  $p$ -norms. To keep the presentation concrete, we first carry out the construction using  $\|\cdot\|_\infty$ , and then explain the modification needed for a general  $p$ -norm.

## 9 Constructing an exotic norm using a Hamel basis

Since the family from the lemma is linearly independent, it can be extended to a Hamel basis  $\mathcal{B}$  of  $C[a, b]$ .

We now define a linear functional on that basis.

**Definition 4.** Define  $L : C[a, b] \rightarrow \mathbb{R}$  on the Hamel basis  $\mathcal{B}$  by prescribing

$$L(e_n) = 0, \quad L(Te_n) = n \quad (n = 0, 1, 2, \dots),$$

and setting

$$L(b) = 0 \quad \text{for every other basis vector } b \in \mathcal{B}.$$

Then extend  $L$  linearly to all of  $C[a, b]$ .

This is well-defined because every element of  $C[a, b]$  is a finite linear combination of basis vectors.

Now define

$$\|f\|_* := \|f\|_\infty + |L(f)|.$$

**Proposition 3.** The formula

$$\|f\|_* := \|f\|_\infty + |L(f)|$$

defines a norm on  $C[a, b]$ .

*Proof.* Nonnegativity and homogeneity are immediate. The triangle inequality follows from the triangle inequality for  $\|\cdot\|_\infty$  and the linearity of  $L$ :

$$\|f + g\|_* = \|f + g\|_\infty + |L(f + g)| \leq \|f\|_\infty + \|g\|_\infty + |L(f)| + |L(g)| = \|f\|_* + \|g\|_*.$$

Finally, if  $\|f\|_* = 0$ , then in particular  $\|f\|_\infty = 0$ , so  $f = 0$ . □

## 10 The counterexample norm based on $\|\cdot\|_\infty$

We now show that the Volterra operator is not continuous with respect to  $\|\cdot\|_*$ .

**Theorem 2.** There exists a norm on  $C[a, b]$  such that the Volterra operator

$$T : C[a, b] \rightarrow C[a, b]$$

is not continuous.

*Proof.* Take the norm  $\|\cdot\|_*$  constructed above.

For each  $n$ , we have  $L(e_n) = 0$  and  $\|e_n\|_\infty = 1$ , so

$$\|e_n\|_* = 1.$$

On the other hand,

$$\|Te_n\|_* = \|Te_n\|_\infty + |L(Te_n)|.$$

Now

$$L(Te_n) = n,$$

and from the explicit formula for  $Te_n$  we get

$$\|Te_n\|_\infty = \frac{b-a}{2n+1}.$$



Therefore

$$\|Te_n\|_* = \frac{b-a}{2n+1} + n.$$

Hence

$$\frac{\|Te_n\|_*}{\|e_n\|_*} = \frac{b-a}{2n+1} + n \longrightarrow \infty.$$

So there cannot exist a constant  $C$  such that

$$\|Tf\|_* \leq C \|f\|_* \quad \text{for all } f \in C[a, b].$$

Thus  $T$  is unbounded, and since  $T$  is linear, it is not continuous.  $\square$

## 11 The same Hamel-basis idea works for every usual $p$ -norm

The preceding construction used the sup norm as the “honest” analytic norm and then added the algebraically defined term  $|L(f)|$ . The same idea works if one starts from any other usual  $p$ -norm on  $C[a, b]$ .

Fix  $1 \leq p < \infty$ . Define

$$u_n(x) := \left( \frac{2np+1}{b-a} \right)^{1/p} \left( \frac{x-a}{b-a} \right)^{2n}.$$

Then a direct calculation shows that

$$\|u_n\|_p^p = \frac{2np+1}{b-a} \int_a^b \left( \frac{x-a}{b-a} \right)^{2np} dx = 1,$$

so

$$\|u_n\|_p = 1.$$

Moreover, each  $u_n$  is just a nonzero scalar multiple of  $e_n$ , so the family

$$\{u_0, u_1, u_2, \dots\} \cup \{Tu_0, Tu_1, Tu_2, \dots\}$$

is still linearly independent. Extend it to a Hamel basis, define a linear functional  $L_p$  by

$$L_p(u_n) = 0, \quad L_p(Tu_n) = n,$$

and let  $L_p$  vanish on all the other basis vectors. Then define

$$\|f\|_{p,*} := \|f\|_p + |L_p(f)|.$$

This is a norm on  $C[a, b]$ , and

$$\|u_n\|_{p,*} = 1, \quad \|Tu_n\|_{p,*} \geq |L_p(Tu_n)| = n.$$

Therefore  $T$  is not continuous with respect to  $\|\cdot\|_{p,*}$ .

So the Hamel-basis construction can be carried out starting from any of the standard norms  $\|\cdot\|_p$  on  $C[a, b]$ , not only from  $\|\cdot\|_\infty$ .

## 12 What this example means

The moral is not that the Volterra operator is badly behaved. Quite the opposite: with respect to all the familiar analytic norms, it is very well behaved.

Rather, the lesson is that in infinite-dimensional spaces there are many norms besides the familiar ones, and some of them can be tailored to force a given linear operator to fail continuity.

The norm  $\|\cdot\|_*$  above is not natural from the viewpoint of analysis. It is built by means of a Hamel basis, which is an algebraic object rather than an analytic one. This is why such counterexamples rarely appear in introductory courses: they are conceptually useful, but they do not reflect the usual geometric or analytic structure of function spaces.

## 13 The space $C_b(I)$ on an arbitrary interval

It is natural to ask what changes if one replaces  $C[a, b]$  by the space

$$C_b(I) := \{f : I \rightarrow \mathbb{R} : f \text{ is continuous and bounded}\}$$

on an arbitrary interval  $I$ .

### Bounded intervals

If  $I = [a, b]$  is a bounded interval, then  $C_b(I) = C[a, b]$ , so everything in this handout applies without change.

### Unbounded intervals

If  $I$  is unbounded, the situation changes immediately. For a fixed base point  $x_0 \in I$ , define formally

$$(Tf)(x) := \int_{x_0}^x f(t) dt.$$

Even when  $f \in C_b(I)$ , the function  $Tf$  need not be bounded. For example, if  $f \equiv 1$ , then

$$(Tf)(x) = x - x_0,$$

which is unbounded on an unbounded interval. Thus, on such an interval, the Volterra operator is generally *not even a map*

$$T : C_b(I) \rightarrow C_b(I).$$

So the basic boundedness question already fails before one asks about continuity.

There is also an important difference for the finite  $p$ -norms. On an unbounded interval, the expression

$$\|f\|_p = \left( \int_I |f(x)|^p dx \right)^{1/p}$$

need not be finite for every bounded continuous function. For instance, the constant function 1 belongs to  $C_b(I)$  but is not integrable when  $I$  has infinite length. Thus, for unbounded intervals,  $\|\cdot\|_p$  with  $p < \infty$  is generally *not* a norm on all of  $C_b(I)$ .

By contrast, the sup norm

$$\|f\|_\infty := \sup_{x \in I} |f(x)|$$

is always a norm on  $C_b(I)$ . On a bounded interval one has the estimate

$$\|Tf\|_\infty \leq |I| \|f\|_\infty,$$

but on an unbounded interval there is no such uniform estimate, because the “length” of the interval is infinite. This is exactly why the example  $f \equiv 1$  causes trouble.

## 14 Summary

We have shown the following.

1. For every  $1 \leq p \leq \infty$ , the Volterra operator satisfies

$$\|Tf\|_p \leq (b - a) \|f\|_p,$$

so it is continuous on  $(C[a, b], \|\cdot\|_p)$ .

2. If one allows arbitrary metrics, then there are trivial counterexamples: for example, if the codomain is given the discrete metric,  $T$  is not continuous.
3. If one insists on a norm, the question becomes subtler. Nevertheless, by using a Hamel basis one can build a norm  $\|\cdot\|_*$  on  $C[a, b]$  such that  $T$  is not continuous.

Thus the correct conclusion is:

The Volterra operator is continuous for all the standard  $p$ -norms on  $C[a, b]$ , but not for all possible norms or metrics.